Advanced Statistics Project

Problem 1

A research laboratory was developing a new compound for the relief of severe cases of hay fever. In an experiment with 36 volunteers, the amounts of the two active ingredients (A & B) in the compound were varied at three levels each. Randomization was used in assigning four volunteers to each of the nine treatments.

Exploratory Data Analysis

	A	B	Volunteer	Relief
0	1	1	1	2.4
1	1	1	2	2.7
2	1	1	3	2.3
3	1	1	4	2.5
4	1	2	1	4.6

Dataset has 4 Variables, A, B, Volunteer and Relief. A, B, and Volunteer are categorical variables and Relief is a float type.

Summary of the Dataset

	A	B	Volunteer	Relief
count	36.000000	36.000000	36.000000	36.000000
mean	2.000000	2.000000	2.500000	7.183333
std	0.828079	0.828079	1.133893	3.272090
min	1.000000	1.000000	1.000000	2.300000
25%	1.000000	1.000000	1.750000	4.675000
50%	2.000000	2.000000	2.500000	6.000000
75%	3.000000	3.000000	3.250000	9.325000
max	3.000000	3.000000	4.000000	13.500000

Information regarding the Data

```
RangeIndex: 36 entries, 0 to 35
Data columns (total 4 columns):
A            36 non-null int64
B            36 non-null int64
Volunteer    36 non-null int64
Relief       36 non-null float64
dtypes: float64(1), int64(3)
```

There are no null values present in the dataset

1.1) State the Null and Alternative Hypothesis for conducting one-way ANOVA for both the variables ‘A’ and ‘B’ individually.

A

- Null Hypothesis - The mean hours of Relief for levels 1,2 and 3 are equal
- Alternative Hypothesis - At Least one of the means are not equal

B

- Null Hypothesis - The mean hours of Relief for levels 1,2 and 3 are equal
- Alternative Hypothesis - At Least one of the means are not equal

1.2) Perform one-way ANOVA for variable ‘A’ with respect to the variable ‘Relief’. State whether the Null Hypothesis is accepted or rejected based on the ANOVA results.

Upon analysis in Python the following ANOVA table was created

	Degrees of Freedom	Sum of squares	Mean of Squares	F	P-value
A	2	220.02	110.01	23.47	0.00000046
Residual	33	154.71	4.69		

Since the P-value is lower than the significant level of 0.05 we reject the Null Hypothesis, thus at least one of the mean hours of Relief is not equal for variable A

1.2) Perform one-way ANOVA for variable ‘A’ with respect to the variable ‘Relief’. State whether the Null Hypothesis is accepted or rejected based on the ANOVA results.

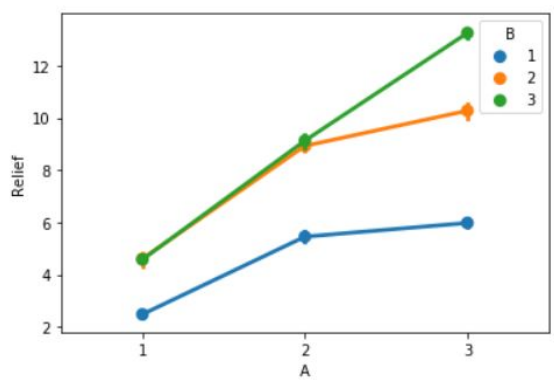
Upon analysis in Python the following ANOVA table was created

	Degrees of Freedom	Sum of Squares	Mean of Squares	F	P-value
B	2	123.66	61.83	8.13	0.00135
	33	251.07	7.61		

Since the P-value is lower than the significant level of 0.05 we reject the Null Hypothesis, thus at least one of the mean hours of Relief is not equal for variable B

1.4) Analyse the effects of one variable on another with the help of an interaction plot.
What is an interaction between two treatments?

The following shows the Interaction Plot drawn between the Variables A, B and relief



In an interaction Plot for ANOVA if the pointplot drawn is a straight line the Means of those treatments are equal. In this case neither of the treatments for variables A and B form a straight line, thus proving the results of the ANOVA table. Although for variable B levels 2 and 3 seem to interact at the same level but divulge at the end of the plot. There is significant evidence to prove that at least one of the means of both variable A and B is not equal.

1.5) Perform a two-way ANOVA based on the different ingredients (variable ‘A’ & ‘B’) with the variable 'Relief' and state your results.

Upon analysis in Python the following ANOVA table was created

	Degrees of Freedom	Sum of Squares	Mean of squares	F	P-value
A	2	220.02	110.01	109.83	8.51e-15
B	2	123.66	61.83	61.73	0.008
Residual	31	31.05	1.00		

Since the P-value is lower than the significant level of 0.05 for both we reject the Null Hypothesis, thus at least one of the mean hours of Relief is not equal

1.6) Mention the business implications of performing ANOVA for this particular case study.

- ANOVA is a test conducted to check the means of more than 2 variables, but it only shows if the means are equal or not for at least one treatment. Further analysis should be done to conclude which mean is equal and which is not to take necessary decisions

- But the existing implications of ANOVA for this case study is that the ANOVA table results conclude that there are differences in the treatments and thus further testing must be done
- ingredient B will be focused on as it has more interaction and possibly provides increased hours of Relief than ingredient A as in the Interaction Plot

Problem 2

The dataset Education - Post 12th Standard.csv is a dataset which contains the names of various colleges. This particular case study is based on various parameters of various institutions. You are expected to do Principal Component Analysis for this case study according to the instructions given in the following rubric.

2.1) Perform Exploratory Data Analysis [both univariate and multivariate analysis to be performed]. The inferences drawn from this should be properly documented.

	0	1	2	3	4
Names	Abilene Christian University	Adelphi University	Adrian College	Agnes Scott College	Alaska Pacific University
Apps	1660	2186	1428	417	193
Accept	1232	1924	1097	349	146
Enroll	721	512	336	137	55
Top10perc	23	16	22	60	16
Top25perc	52	29	50	89	44
F.Undergrad	2885	2683	1036	510	249
P.Undergrad	537	1227	99	63	869
Outstate	7440	12280	11250	12960	7560
Room.Board	3300	6450	3750	5450	4120
Books	450	750	400	450	800
Personal	2200	1500	1165	875	1500
PhD	70	29	53	92	76
Terminal	78	30	66	97	72
S.F.Ratio	18.1	12.2	12.9	7.7	11.9
perc.alumni	12	16	30	37	2
Expend	7041	10527	8735	19016	10922
Grad.Rate	60	56	54	59	15

There are 18 variables in this Dataset

Information regarding the Data

```
RangeIndex: 777 entries, 0 to 776
Data columns (total 18 columns):
Names          777 non-null object
Apps           777 non-null int64
Accept         777 non-null int64
Enroll         777 non-null int64
Top10perc      777 non-null int64
Top25perc      777 non-null int64
F.Undergrad    777 non-null int64
P.Undergrad    777 non-null int64
Outstate       777 non-null int64
Room.Board     777 non-null int64
Books          777 non-null int64
Personal       777 non-null int64
PhD            777 non-null int64
Terminal       777 non-null int64
S.F.Ratio      777 non-null float64
perc.alumni    777 non-null int64
Expend         777 non-null int64
Grad.Rate      777 non-null int64
dtypes: float64(1), int64(16), object(1)
```

There is one Object type, 16 Integer type and 1 float type

Summary of Data

	count	mean	std	min	25%	50%	75%	max
Apps	777.0	3001.638353	3870.201484	81.0	776.0	1558.0	3624.0	48094.0
Accept	777.0	2018.804376	2451.113971	72.0	604.0	1110.0	2424.0	26330.0
Enroll	777.0	779.972973	929.176190	35.0	242.0	434.0	902.0	6392.0
Top10perc	777.0	27.558559	17.640364	1.0	15.0	23.0	35.0	96.0
Top25perc	777.0	55.796654	19.804778	9.0	41.0	54.0	69.0	100.0
F.Undergrad	777.0	3699.907336	4850.420531	139.0	992.0	1707.0	4005.0	31643.0
P.Undergrad	777.0	855.298584	1522.431887	1.0	95.0	353.0	967.0	21836.0
Outstate	777.0	10440.669241	4023.016484	2340.0	7320.0	9990.0	12925.0	21700.0
Room.Board	777.0	4357.526384	1096.696416	1780.0	3597.0	4200.0	5050.0	8124.0
Books	777.0	549.380952	165.105360	96.0	470.0	500.0	600.0	2340.0
Personal	777.0	1340.642214	677.071454	250.0	850.0	1200.0	1700.0	6800.0
PhD	777.0	72.660232	16.328155	8.0	62.0	75.0	85.0	103.0
Terminal	777.0	79.702703	14.722359	24.0	71.0	82.0	92.0	100.0
S.F.Ratio	777.0	14.089704	3.958349	2.5	11.5	13.6	16.5	39.8
perc.alumni	777.0	22.743887	12.391801	0.0	13.0	21.0	31.0	64.0
Expend	777.0	9660.171171	5221.768440	3186.0	6751.0	8377.0	10830.0	56233.0
Grad.Rate	777.0	65.463320	17.177710	10.0	53.0	65.0	78.0	118.0

- The difference between the 3rd Quartile and max is high denoting there is a good chance of outliers present in the Data
- Colleges receive 3000 applications on an average
- There are more full time grads than part time
- The average Graduation Rate is 65%

Checking for Duplicates

```
dups = df.duplicated()
print('Number of duplicate rows = %d' % (dups.sum()))

df[dups]
```

```
Number of duplicate rows = 0
```

There are no duplicates in the Dataset

Checking for Null values

```
df.isnull().sum()
```

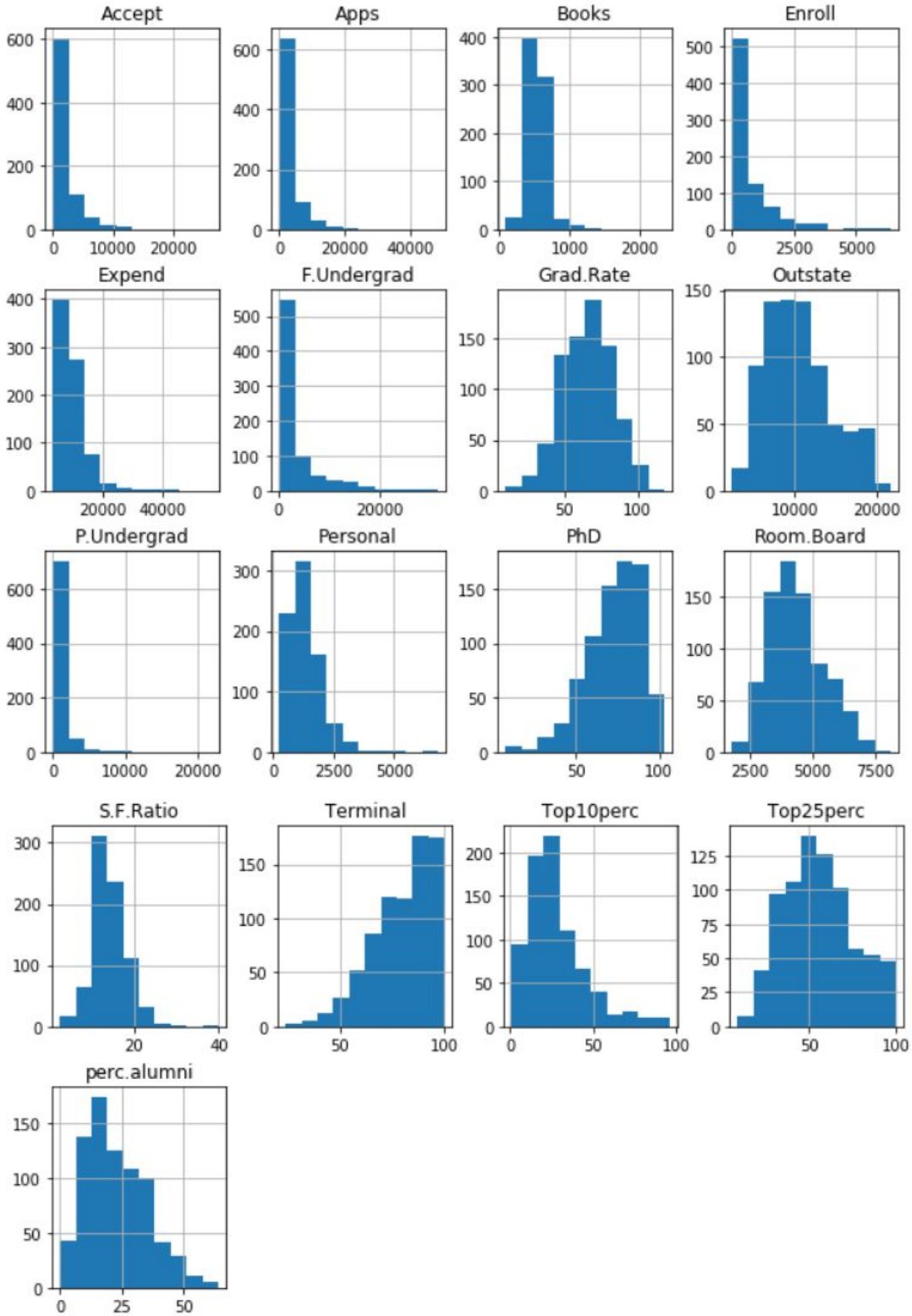
```
Names          0
Apps           0
Accept         0
Enroll         0
Top10perc      0
Top25perc      0
F.Undergrad    0
P.Undergrad    0
Outstate       0
Room.Board     0
Books          0
Personal       0
PhD            0
Terminal       0
S.F.Ratio      0
perc.alumni    0
Expend         0
Grad.Rate      0
dtype: int64
```

There are no Null values in the Dataset

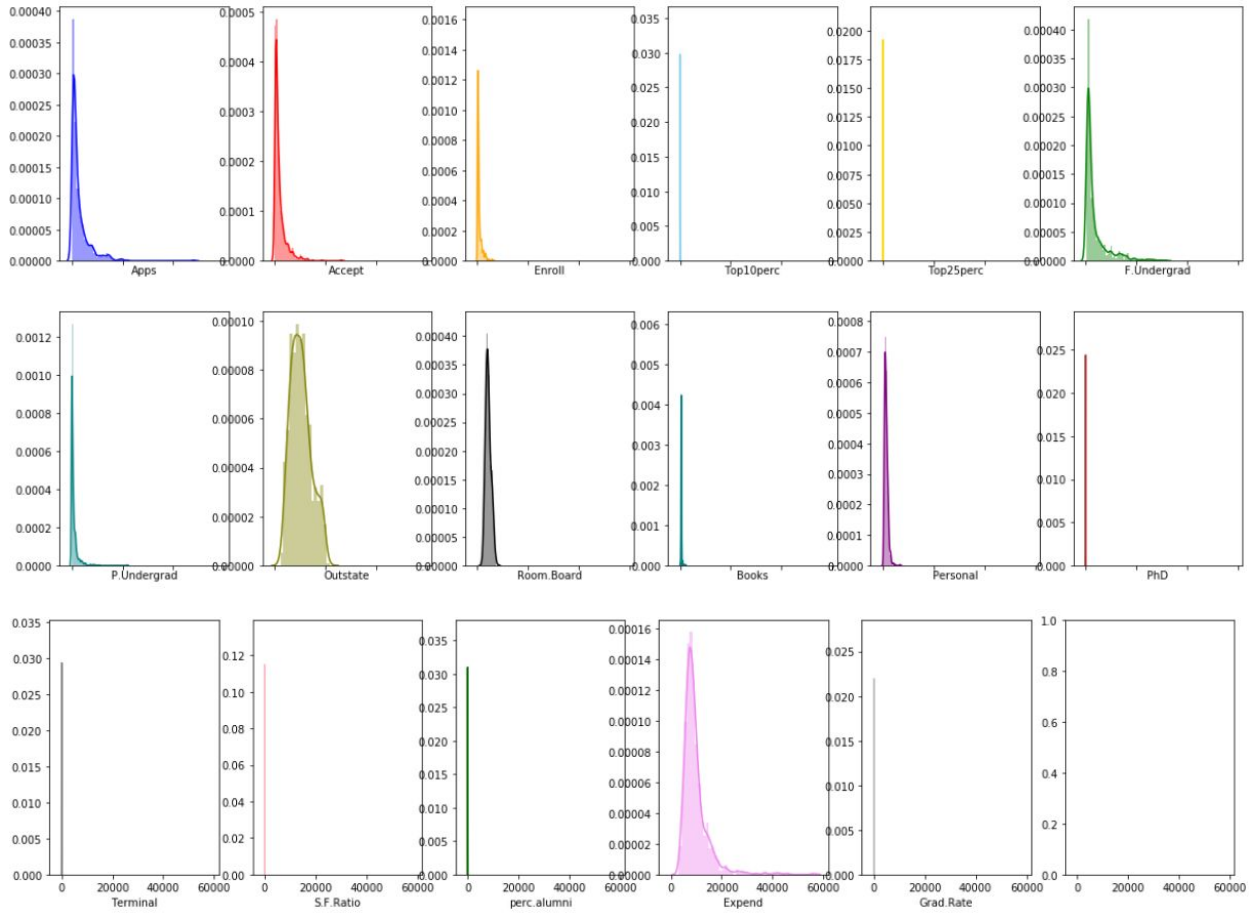
Univariate Analysis

Histograms were drawn for each variable of the Dataset and the following inferences were made

- Most of the Colleges receive roughly 5000 applications although there a select few who receive from 10000 to 20000 applications
- The cost of books in most colleges is below 1000
- More than 10000 students prefer to study out of state
- The personal expenditure of most students is roughly below 2500
- There is a significant amount of increase in Terminal Degree which each bar
- The most common Student Faculty ratio is between 10 and 25
- The percentage of top25% is high at 50


Accept
Apps
Books
Enroll
Expend
F.Undergrad
Grad.Rate
Outstate
P.Undergrad
Personal
PhD
Room.Board
S.F.Ratio
Terminal
Top10perc
Top25perc
perc.alumni

Distance Plot



- Application, Accept, F.UnderGrad, Expenditure, Enroll, P.UnderGrad, Top 10perc, perc.alumni and Personal are all skewed to the right
- Outstate, Room.Board, Top 25perc, Books, S.F Ratio and Grad rate are normally distributed
- PhD, Terminal are skewed to the left

Multivariate Analysis

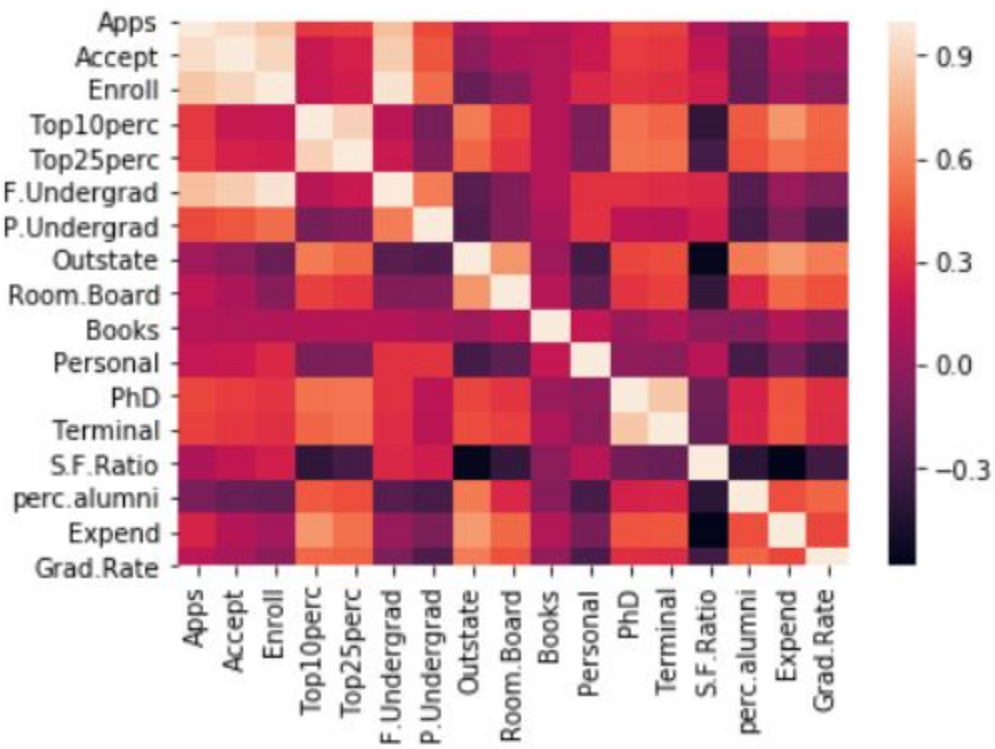
Pairplot was drawn to analyze the behavior between each variable with one another.

- Most of the variables have a negative correlation with one another will some show positive

- Terminal Degree, books and P.Undergrad is high at the beginning but decreases when the other variable increases
- The S.F Ratio and Grad.Rate show neutral behavior with all variables
- Applications has a positive correlation with accept and enroll but neutral behavior with others
- The rest of the variables are either neutral or negatively behaving

*Refer Python file for visualization

Correlation between the variables was calculated and a heatmap was drawn based on the results



- Percentage of top 10 and 25 percent is highly correlated with S.F Ratio
- Outstate, Room.Board, perc.alumni, Expend and Grad Rate are also highly correlated with S.F Ratio
- There is medium to high correlation between Top 10 and 25% with Part time Grads

- Applications, Accept and Enroll have very low correlation with each other

  Outlier checking before Scaling

  The Dataset was checked for outliers using boxplot
  - The result showed significant outliers in most of the variables
  - There are no outliers in Top 25perc
  - There are fewer outliers in Outstate, Perc.alumni and Grad.Rate

*Refer Python file for visualization

2.2) Scale the variables and write the inference for using the type of scaling function for this case study.

	0	1	2	3	4
Apps	-0.346882	-0.210884	-0.406866	-0.668261	-0.726176
Accept	-0.321205	-0.038703	-0.376318	-0.681682	-0.764555
Enroll	-0.063509	-0.288584	-0.478121	-0.692427	-0.780735
Top10perc	-0.258583	-0.655656	-0.315307	1.840231	-0.655656
Top25perc	-0.191827	-1.353911	-0.292878	1.677612	-0.596031
F.Undergrad	-0.168116	-0.209788	-0.549565	-0.658079	-0.711924
P.Undergrad	-0.209207	0.244307	-0.497090	-0.520752	0.009005
Outstate	-0.746356	0.457496	0.201305	0.626633	-0.716508
Room.Board	-0.964905	1.909208	-0.554317	0.996791	-0.216723
Books	-0.602312	1.215880	-0.905344	-0.602312	1.518912
Personal	1.270045	0.235515	-0.259582	-0.688173	0.235515
PhD	-0.163028	-2.675646	-1.204845	1.185206	0.204672
Terminal	-0.115729	-3.378176	-0.931341	1.175657	-0.523535
S.F.Ratio	1.013776	-0.477704	-0.300749	-1.615274	-0.553542
perc.alumni	-0.867574	-0.544572	0.585935	1.151188	-1.675079
Expend	-0.501910	0.166110	-0.177290	1.792851	0.241803
Grad.Rate	-0.318252	-0.551262	-0.667767	-0.376504	-2.939613

- Z score was applied to scale the variables, although both applying Z score and Standard Scaler gave the same results Z score was used because in Standard Scaler each Variable had to be called out individually. Using Z score is a more efficient method compared to Standard Scaler as it is time saving.

2.3) Comment on the comparison between covariance and the correlation matrix.

- Covariance shows how two variables vary together while correlation shows if one variable increases or decreases when the other increases
- When compared Covariance focuses on the variance between two variables while Correlation focuses on direction of movement

2.4) Check the dataset for outliers before and after scaling. Draw your inferences from this exercise.

The Dataset was checked for outliers after scaling the data which showed an increase in outliers but the Quartile values were all similar for the variables. Scaling does not treat outliers, it only standardizes the data where all variables have similarities making it easier to analyze.
In this case the outliers may have increased but the boxplots were all at similar values thus standardization was achieved.

2.5) Build the covariance matrix, eigenvalues and eigenvector.

```
 Eigen Values
%s [5.45052162 4.48360686 1.17466761 1.00820573 0.93423123 0.84849117
 0.6057878  0.58787222 0.53061262 0.4043029  0.02302787 0.03672545
 0.31344588 0.08802464 0.1439785  0.16779415 0.22061096]
```

As shown above the Eigenvectors have been calculated from the covariance matrix

*Refer Python file for Covariance Matrix and Eigenvalues

2.6) Write the explicit form of the first PC (in terms of EigenVectors)

	PC-1
Apps	0.248766
Accept	0.207602
Enroll	0.176304
Top10perc	0.354274
Top25perc	0.344001
F.Undergrad	0.154641
P.Undergrad	0.026443
Outstate	0.294736
Room.Board	0.249030
Books	0.064758
Personal	-0.042529
PhD	0.318313
Terminal	0.317056
S.F.Ratio	-0.176958
perc.alumni	0.205082
Expend	0.318909
Grad.Rate	0.252316

2.7) Discuss the cumulative values of the eigenvalues. How does it help you to decide on the optimum number of principal components? What do the eigenvectors indicate?
Perform PCA and export the data of the Principal Component scores into a data frame.

The Cumulative Variance explained shows percentage of data summed up in each principal component. With each principal component the Cumulative Variance keeps increasing till it reaches 100%. The purpose of PCA is dimension reduction. Cumulative Variance helps in deciding the optimal number of components which will cover at least 70 to 95% of the data.

The Eigenvectors indicate the direction of the new feature space for the reduced data.

PCA was performed and the Principal Component Scores were exported to the Dataset as a separate column “PCA_score”

2.8) Mention the business implication of using the Principal Component Analysis for this case study.

- PCA is used for dimensionality reduction, in this case there were 17 variables or parameters regarding each of those colleges.
- The results can be used to find out which of those 17 variables is important in comparing the colleges.
- Although not analyzed PCA can also be used to rank the best college using the PCA scores